

Johnathan Aguilar

Education

University of Massachusetts Amherst
MS in Computer Science (Data Science Concentration)
◦ GPA: 3.90/4.00

Sept 2026 – Present

California State University, Northridge
BS in Computer Science, Minor in Data Science
◦ GPA: 3.95/4.0

Aug 2023 – May 2026

Research Interest

Multimodal AI, Computer Vision, Foundation Models, Embodied AI, World Models, Spatiotemporal Modeling, Remote Sensing, Knowledge Graphs, and Real-World AI Systems

Research Experience

AI Research Intern

Waste Management & Autonomy Research Center For STEAHM

Sun Valley, CA

Sept 2025 – Present

- Lead an undergraduate team in collaboration with a PhD student and a faculty advisor, coordinating research tasks and ensuring alignment with project goals.
- Contributed to the implementation of a Neo4j knowledge graph enabling multimodal, context-aware reasoning for an intelligent agent supporting facility safety and operations.
- Developed a Gen-AI pipeline to convert semi-structured handwritten reports into graph representations.
- Supported research on hybrid graph-vector knowledge representations, demonstrating their potential to enhance reasoning and multimodal understanding in complex environments.

Undergraduate Researcher

California State University, Northridge - Department of Computer Science

Northridge, CA

Oct 2025 – Present

- Conducted spatio-temporal analysis of Level-4 Sea Surface Temperature (SST) data from NASA PO.DAAC in the Caribbean near the Archipelago of San Andrés and around Hawaii's main island as part of a NASA JPL-sponsored research project.
- Performed Empirical Orthogonal Function (EOF) analysis and time series clustering to identify locations sharing similar temperature patterns over time, enabling the detection of thermally distinct zones across the study area.
- Performed statistical and exploratory data analysis to reveal seasonal spatio-temporal dynamics and contribute insights toward understanding coastal resilience under heatwave conditions.

Research Intern

NASA Jet Propulsion Laboratory (JPL)

Pasadena, CA

June 2026 – Present

- Develop deep learning models for segmentation of oceanographic phenomena using SWOT (Surface Water and Ocean Topography) remote sensing data
- Design data pipelines for preprocessing, tiling, and labeling large-scale geospatial datasets for supervised and self-supervised learning
- Analyze model performance using metrics such as IoU, precision, recall, and F1-score, and optimize architectures for spatiotemporal generalization

Publications

Multimodal Generative AI Agent for Event-Based Proactive Maintenance,
CIRP Annals – Manufacturing Technology (2026, under review)

Bingbing Li, Xuan Liu, **Johnathan Aguilar**, Hossein Khandan, John Ahmet Erkoyuncu, Chris Yuan

Presentation & Posters

Statistical Analysis of MUR Sea Surface Temperature: Approaches to Thermal Refugia Detection in the Seaflower MPA

Feb 2026

Lightning talk presentation at the LA Geospatial Summit, University of Southern California.

Authors: **Johnathan Aguilar**

Honors and Achievements

Autonomy Research Center For STEAHM Fellowship	2025 - 2026
California State University, Northridge Dean's List, 5 semesters	2023 - 2025
Moorpark College Dean's List	2022

Technologies

Languages:	Python, R, C/C++, Java, JavaScript, TypeScript, SQL, HTML, CSS
Libraries/framework:	React, Express.js, pydoc, pytest, NumPy, Pandas, Matplotlib, scikit-learn, XGBoost
Technologies:	AWS (EC2, RDS), DigitalOcean, Node.js, Maven, Docker, Git, GitHub, JIRA, Jupyter Notebook, Anaconda
Systems	Linux, Windows